Pokhara University
Faculty of Science and Technology

Course No.: Full marks: 100

Course title: Information System Audit (3-1-0) Pass marks: 45

Nature of the course: Theory Total Lectures: 45 hrs.

Level: Bachelors Program: BE

1. Course Description

The Information System Audit course for undergraduate students provides an in-depth understanding of auditing principles, methodologies, and best practices for assessing the security, integrity, and compliance of information systems. The course covers risk assessment, control frameworks, IT governance, regulatory compliance, and emerging threats in cybersecurity. Students will gain hands-on experience in auditing IT infrastructure, applications, and processes through case studies and practical exercises. This course covers different concepts of Information Systems Auditing including basics, hardware and software security issues, information systems audit and conducting IS audit, risk-based systems audit, business continuity and disaster, auditing in the ICT environment, and security testing.

2. General Objectives

- To develop a foundational understanding of information systems auditing principles, processes, and methodologies.
- To explore risk-based auditing approaches within ICT environments.
- To examine the role of Governance, Risk, and Compliance (GRC) in information systems auditing.
- To understand the importance of business continuity and disaster recovery in audit planning.
- To gain familiarity with security testing tools, vulnerability assessment, and penetration testing techniques.
- To analyze emerging trends and challenges in the field of information systems auditing.

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3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies and Group Workshops

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4. Contents in Detail.

Specific Objectives	Contents
Acquire knowledge of the fundamental concepts of Information System Audit and IT Audit Frameworks, with an emphasis on widely recognized standards such as ISO 27001, NIST, and MITRE.	**Unit 1: Overview of Systems Audit (8 Hrs.)** 1.1 Information Systems Audit and Information Systems Auditor 1.2 Legal Requirements of an Information Systems Audit 1.3 Systems Environment and Information Systems Audit 1.4 Information Systems Assets and Classification of Controls 1.5 Information Systems Audit Coverage 1.6 IT Audit Framework, ISO 27001, NIST Cyber Security Framework and MITRE
Gain knowledge of the key concepts related to hardware and software security issues encountered during an Information System Audit.	**Unit 2: Hardware and Software Security Issues (8 Hrs.)** 2.1 Hardware Security Objective 2.2 Peripheral Devices and Storage Media 2.3 Authentication Devices 2.4 Hardware Acquisition, Hardware Maintenance and Management of Obsolescence 2.5 Disposal of Equipment; Problem Management; Change Management 2.6 Network and Communication Issues. 2.7 Overview of Types of Software; Elements of Software Security 2.8 Control Issues during Installation and Maintenance
Gain knowledge of the requirements of Information System Audit in relation to system controls, log management, and evidence collection processes.	**Unit 3: Information Systems Audit Requirements (8 Hrs.)** 3.1 Information Systems Control Objectives; Information Systems Audit Objectives; 3.2 System Effectiveness and Efficiency 3.3 Information Systems Abuse 3.4 Asset Safeguarding Objective and Process 3.5 Evidence Collection and Evaluation 3.6 Logs and Audit Trails as Evidence 3.7 IT Audit Standard and Regulatory Requirements
Gain knowledge of the concept of Information System Auditing, including its purpose, scope, and role in evaluating and ensuring the effectiveness of IT controls and processes.	**Unit 4: Conducting an Information Systems Audit (10 Hrs.)** 4.1 Audit Program and Audit Plan 4.2 Audit Procedures and Approaches 4.3 System Understanding and Review 4.4 Compliance Reviews and Tests 4.5 Substantive Reviews and Tests 4.6 Audit Tools and Techniques 4.7 Sampling Techniques 4.8 Audit Questionnaire; Audit Documentation; Audit Report 4.9 Auditing Approaches; Sample Audit Work-Planning Memo

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	4.10 Sample Audit Work Process Flow 4.11 Conducting a Risk-Based Information Systems Audit 4.12 Risk Assessment and Risk Management Strategy
Gain knowledge of the concept of Business Continuity and Disaster Recovery Planning within the context of Information System Audits, focusing on strategies to ensure organizational resilience and recovery in the event of disruptions.	**Unit 5: Business Continuity and Disaster Recovery Plan (6 Hrs.)** 5.1 Business Continuity and Disaster Recovery Process 5.2 Business Impact Analysis; Incident Response Plan 5.3 Disaster Recovery Plan 5.4 Types of Disaster Recovery Plans 5.5 Emergency Preparedness Audit Checklist 5.6 Business Continuity Strategies 5.7 Business Resumption Plan Audit Checklist 5.8 Recovery Procedures Testing Checklist; Plan Maintenance Checklist.
Apply the concept of VAPT and Emerging Concepts	**Unit 6: Security Testing and Cloud Computing Audit (5 Hrs.)** 6.1 Cybersecurity 6.2 Vulnerability Assessment and Penetration Testing (VAPT) 6.3 Security Testing Tools 6.4 Emerging Concepts in Information System Audit

5. Tutorial and Group Workshops (15 Hours)

The tutorial and group workshop should cover the following works:

SN	Group Workshops
1.	Information Security Gap Assessment through ISO 27001
2.	Cybersecurity Maturity Assessment through NIST Cybersecurity Framework using CISA-CSET or similar tools
3.	Conduct Information System Audit and Reporting
4.	Conducting Risk Assessment
5	Case Study: GRC Cybersecurity Engine

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6. **Evaluation system and Students' Responsibilities**

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment and group workshop. The internal evaluation scheme for this course is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory			Semester-End examination	50
Attendance & Class Participation	10%	5		
Assignments	10%	5		
Presentations/Quizzes	10%	5		
Internal Assessment	50%	25		
Group Workshops	20%	10		
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

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7. Prescribed Books and References

References:

1. Hall, J. A. (2020). *Information Technology Auditing and Assurance* (5th ed.). Cengage Learning
2. Moeller, R. (2022). *IT Audit, Control, and Security* (3rd ed.). Wiley.
3. ISACA. (2024). *CISA Review Manual.* ISACA.
4. ISACA. (2022). *COBIT 2019 Framework: Governance and Management of Enterprise IT.* ISACA
5. Veena Hingarh and Arif Ahmed (2013), *Understanding and Conducting Information Systems Auditing*, Wiley
6. Jack J. Champlain (2003), *Auditing Information Systems*, 2nd Edition, Wiley
7. Richard Cascarino (2007), *Auditor's Guide to Information Systems Auditing*, Wiley
8. ISACA Journals
9. ISACA IT Audit Frameworks
10. NIST Special Publications

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